

community. It enables us to be more lean, agile and secure and, hence, move faster in the competitive world," says Kumar.

'More exciting than challenging'

Although shifting to open source turned out to be a fruitful decision for Airtel, the transition was not a breeze. However, Kumar terms this transition "more exciting than challenging."

The main challenge that Kumar and his team faced while switching to open source was to bridge the knowledge gap. Airtel preferred reskilling the existing team and hired fresh talent only to make the transition smoother.

"Having the right team is very important. The engineering team does look for new co-workers who enjoy doing their work and are curious to discover the given solutions. At times, we ask them to code during interviews just to see how they think," Kumar says.

Kumar adds that his engineering crew picks solutions that are tried and tested by various large- and small-scale organisations, have a large community base, elaborate documentation and are available without any restrictive or expensive licensing terms and conditions. Moreover, the team follows the agile development methodology and believes in prototyping technologies before their final deployment.

"Following agile development methodology helped us," says Kumar, adding, "With rapid prototyping, we can judge the fitness of a technology for an application."



Anant Kumar, head of engineering, Bharti Airtel, exploring new open source developments with his team

Six major projects that use open source technologies

Airtel has as many as six major projects based on open source solutions. There is the Mitra app for retailers to acquire new customers and reverse wrongly made recharges; the Fibre



Anant Kumar, general manager and head of product engineering, Airtel

Airtel's open source based projects

EeCAF: An app to completely digitise customer acquisition using the Aadhaar number and biometric fingerprint, hence reducing activation TAT (turnaround time) from a few days to a few minutes.

Mitra app: An app empowering retailers to acquire new customers, reverse wrong recharges and keep them up-to-date with their account and services -- ensuring complete synergy between the company's and retailer's achievements.

Decision Tree: A one-stop-shop for call centre employees, which reduces the average call handling and training time considerably.

Fibre Force app: This app enables NMT engineers to measure a patroller's performance for inter-city and intra-city fibre maintenance. It helps to improve operational efficiency and also increase GIS accuracy in the long run. This will also ensure that risk-prone areas are proactively allocated additional resources.

Hive: A social networking app for Airtel employees, which enriches internal communication.

Project Leap: An app that provides an aggregate experience metric, thus enabling the planning for future networks as well as resolving existing network issues.

Force app to help NMT engineers measure a patroller's performance; the Decision Tree app for call centre employees; the Hive social network app and the Project Leap open network initiative. Also, the telecom giant has recently developed its eCAF (electronic customer acquisition form) to digitise customer acquisition using Aadhaar biometric data.

The development of eCAF was aimed at offering instant activation of mobile connections and to enable paperless processes for account activation. "We were getting almost